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The Variational Quantum Eigensolver

From molecular Hamiltonians to a CUDA-Q implementation — explained step by step

CUDA-Q Quantum Algorithm Benchmarking Suite

Companion lecture to `VQE.ipynb`

Quantum results in these slides were computed with NVIDIA CUDA-Q 0.16 (`qpp-cpu`) using the kernels and Hamiltonians of `VQE.ipynb`, and checked against NumPy; chemistry data with PySCF + OpenFermion.

What you will learn today

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1. Eigenvalue problems and why ground states matter
2. Quantum chemistry: Schrödinger equation, molecular Hamiltonians, second quantisation
3. From fermions to qubits: Jordan–Wigner, Bravyi–Kitaev
4. Qubit toolkit: Pauli strings, rotations, measurement
5. The **variational principle** and the VQE loop
6. Ansatz design: Hartree–Fock, **UCC**, hardware-efficient
7. Worked examples: H_2 and the Ising chain
8. The CUDA-Q implementation in `VQE.ipynb`, line by line
9. Results, shots, noise, limitations and current research

You only need

- matrices, eigenvalues and eigenvectors
- complex numbers
- basic calculus (a minimum where $f' = 0$)

Three kinds of statements

General theory, **the notebook's implementation**, and **illustrative examples** (e.g. LiH) are marked as such.

The eigenvalue problem

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Definition

For a square matrix H: find numbers E (eigenvalues) and non-zero vectors $|\psi\rangle$ (eigenvectors) with $H|\psi\rangle = E|\psi\rangle$.

Hermitian matrices ($H = H^\dagger$):

- all eigenvalues are **real**: $E = \langle\psi|H|\psi\rangle^* = E^*$
- eigenvectors of different eigenvalues are **orthogonal**
- an orthonormal eigenbasis exists (spectral theorem): $H = \sum_k E_k |E_k\rangle\langle E_k|$

Example

$H = \left(\begin{smallmatrix} 1 & 2 \\ 2 & -2 \end{smallmatrix}\right)$:
$\det(H - \lambda I) = \lambda^2 + \lambda - 6 = 0$
$\Rightarrow E_0 = -3$, $|E_0\rangle = \frac{1}{\sqrt{5}}(1, -2)$;
$E_1 = 2$.

The smallest eigenvalue E_0 is the **ground-state energy**; VQE estimates it.

Classical eigenvalue computation and its limit

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- Dense diagonalisation: $O(D^3)$ time, $O(D^2)$ memory for a $D \times D$ matrix.
- Sparse iterative methods (Lanczos, `scipy.sparse.linalg.eigsh`): only the lowest few eigenvalues, cost $\approx$ matrix–vector products.
- For n qubits $D = 2^n$: every added qubit doubles the vector.

In `VQE.ipynb`

Exact references: sparse diagonalisation up to `REFERENCE_MAX_QUBITS = 18`; the Ising chain has an exact free-fermion formula for any N.

n	$D = 2^n$	vector (fp64)
4	16	256 B
15	32 768	524 kB
30	1.1×10^9	17 GB
50	1.1×10^{15}	18 PB

Rayleigh quotient: $R(v) = \frac{v^\dagger H v}{v^\dagger v} \geq E_0$ for every $v \neq 0$ — the seed of the variational method.

The Schrödinger equation

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Time-dependent

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$$

Solution: $|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$ (unitary evolution).

Time-independent

$$H |\psi\rangle = E |\psi\rangle$$

Eigenstates are **stationary**: $|\psi(t)\rangle = e^{-iEt/\hbar} |\psi\rangle$ — only a global phase changes.

- H: Hamiltonian = total-energy operator.
- $E_0 < E_1 \leq E_2 \leq \ldots$: allowed energies. E_0 = **ground state**, the others **excited states**.
- At low temperature a molecule sits in its ground state: E_0 decides bond lengths, bond energies and reaction energies.
- Energy differences between states give spectra (colours, absorption lines).

The molecular Hamiltonian

For electrons i, j (positions $\mathbf{r}_i$) and nuclei A, B (charges Z_A, positions $\mathbf{R}_A$), in atomic units:

$$H = \underbrace{-\sum_i \tfrac{1}{2}\nabla_i^2}_{\text{kinetic}} \underbrace{-\sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}}_{\text{electron–nucleus}} + \underbrace{\sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{electron–electron}} + \underbrace{\sum_{A<B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}}_{V_{NN}}$$

Born–Oppenheimer approximation

Nuclei are $\sim 2000\times$ heavier: fix them. V_{NN} becomes a *constant*; solve for the electrons at every geometry $\Rightarrow$ potential energy surface.

Units

1 Hartree (Ha) $= 27.211$ eV $= 627.51$ kcal/mol.

Chemical accuracy: 1 kcal/mol $= 1.6$ mHa — the target for useful predictions.

Electronic structure: orbitals, determinants, correlation

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- Choose a finite **basis** of N_{orb} spatial orbitals (e.g. STO-3G: one function per occupied atomic shell).
- Each holds spin α and β: $N = 2N_{\mathrm{orb}}$ **spin-orbitals**.
- A **Slater determinant** = a choice of N_e occupied spin-orbitals (antisymmetrised product).
- **Hartree–Fock**: best single determinant. **FCI**: exact diagonalisation over all $\binom{N}{N_e}$ determinants.

H_2, STO-3G (the notebook's molecule)

2 spatial orbitals σ_g, $\sigma_u \Rightarrow$ 4 spin-orbitals, 2 electrons, $\binom{4}{2} = 6$ determinants.

$E_{\mathrm{HF}} = -1.116759$ Ha,
$E_{\mathrm{FCI}} = -1.137284$ Ha
correlation energy
$E_{\mathrm{FCI}} - E_{\mathrm{HF}} = -20.5$ mHa.

Classical methods: HF (N^4), CCSD(T) (N^7), DMRG, FCI ($\binom{N}{N_e}$). Strong correlation (bond breaking, transition metals) defeats the cheap ones.

Second quantisation

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Creation $a_p^\dagger$ and annihilation a_p operators fill or empty spin-orbital p:

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \qquad \{a_p, a_q\} = 0$$

- $(a_p^\dagger)^2 = 0$: Pauli exclusion
- $n_p = a_p^\dagger a_p$ has eigenvalues 0, 1
- swapping two operators gives a minus sign: **fermions**

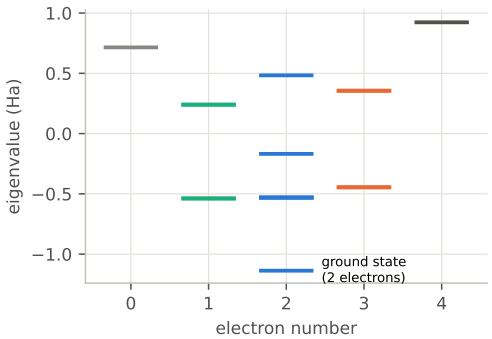
Electronic Hamiltonian

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + E_{\text{const}}$$

h_{pq}: one-electron integrals (kinetic + nuclear attraction), h_{pqrs}: electron repulsion integrals, computed classically (PySCF).

Number of terms $O(N^4)$ — polynomial, although the state space is 2^N.

Ground and excited states of H_2



- All 16 eigenvalues of the notebook's 4-qubit H_2 Hamiltonian, sorted by electron number.
- The global minimum -1.137284 Ha has exactly **2 electrons** — the neutral molecule.
- Exact ground state: $0.9937\,|1100\rangle - 0.1125\,|0011\rangle$, i.e. 98.7 % Hartree–Fock, 1.3 % doubly excited.
- Excited states are ≥ 0.53 Ha higher: VQE only needs the bottom.

Jordan–Wigner transformation

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One qubit per spin-orbital; $|1\rangle$ = occupied, $|0\rangle$ = empty.

Mapping

$$a_j^\dagger = Z_0 Z_1 \cdots Z_{j-1}\, \tfrac{1}{2}(X_j - iY_j), \qquad n_j = \tfrac{1}{2}(I - Z_j)$$

- $\frac{1}{2}(X - iY) = |1\rangle\langle 0|$ fills the orbital.
- The **Z-string** supplies the sign $(-1)^{n_0+\cdots+n_{j-1}}$ that makes operators on different orbitals *anti*commute.
- Example: $a_2^\dagger = \frac{1}{2} Z_0 Z_1 (X_2 - iY_2)$ on 4 qubits.

Hopping term

$a_0^\dagger a_1 + a_1^\dagger a_0 = \frac{1}{2}(X_0 X_1 + Y_0 Y_1)$

Derivation: $a_1 = Z_0 \frac{1}{2}(X_1 + iY_1)$, $(X - iY)Z = X - iY$, so $a_0^\dagger a_1 = \frac{1}{4}(X_0 - iY_0)(X_1 + iY_1)$; adding the conjugate cancels the imaginary parts.

Weight of $a_j^\dagger$: $j + 1$, i.e. $O(N)$.

Bravyi–Kitaev, parity mapping and tapering

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- **Parity mapping**: qubit j stores $n_0 \oplus \cdots \oplus n_j$; occupations become non-local instead.
- **Bravyi–Kitaev**: qubits store partial sums on a binary tree: $b = Bn \bmod 2$. Both occupation and parity need only $O(\log N)$ qubits. E.g. $a_7^\dagger$ on 8 modes: weight 8 (JW) vs. 4 (BK).
- **Tapering**: $\mathbb{Z}_2$ symmetries (electron-number and spin parity) remove qubits exactly; H_2: $4 \to 2$ qubits.

BK on 4 modes

$$B = \begin{pmatrix} 1&0&0&0 \\ 1&1&0&0 \\ 0&0&1&0 \\ 1&1&1&1 \end{pmatrix}$$

HF occupation $n = (1,1,0,0) \to b = (1,0,0,0)$: the same physical state is $|1100\rangle$ in JW but $|1000\rangle$ in BK (checked: $\langle 1000|H_{\mathrm{BK}}|1000\rangle = E_{\mathrm{HF}}$).

`VQE.ipynb` uses neither mapping explicitly: its H_2 coefficients are already Jordan–Wigner.

The notebook's H_2 qubit Hamiltonian

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Pauli string	coefficient (Ha)
IIII	−0.097066
ZIII, IZII	+0.171413
IIZI, IIIZ	−0.223432
ZZII	+0.168689
ZIZI, IZIZ	+0.120625
ZIIZ, IZZI	+0.165928
IIZZ	+0.174413
XXYY, YYXX	−0.045303
XYYX, YXXY	+0.045303

- 15 terms; qubits = spin-orbitals $(\sigma_g\alpha, \sigma_g\beta, \sigma_u\alpha, \sigma_u\beta)$, STO-3G, Jordan–Wigner.
- IIII contains the nuclear repulsion and other constants.
- Z terms: orbital energies and Coulomb/exchange between occupations. XY terms: the double excitation $|1100\rangle \leftrightarrow |0011\rangle$.

Verified provenance

PySCF + OpenFermion reproduce all 15 coefficients to $3e-15$ at bond length $R = 0.740$ **Å**. (The notebook's text originally said 0.7414 Å, where they would differ by up to 1.8 mHa; corrected.)

Walk-through: the Hartree–Fock energy term by term

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term	c_k	$\langle 1100 \vert P_k \vert 1100 \rangle$	contribution
IIII	-0.097066	$+1$	-0.097066
ZIII, IZII	$+0.171413$	-1	-0.342826
IIZI, IIIZ	-0.223432	$+1$	-0.446863
ZZII	$+0.168689$	$+1$	$+0.168689$
ZIZI, IZIZ	$+0.120625$	-1	-0.241250
ZIIZ, IZZI	$+0.165928$	-1	-0.331856
IIZZ	$+0.174413$	$+1$	$+0.174413$
4 exchange terms	± 0.045303	0	0
$E_{\mathrm{HF}} = \langle 1100 \vert H \vert 1100 \rangle$			-1.116759

- On a basis state $|z\rangle$, Z_j gives $+1$ for an empty orbital ($z_j = 0$) and -1 for an occupied one.
- A product $Z_i Z_j$ gives the product of the signs.
- Strings containing X or Y flip bits, so their diagonal element is 0.
- The sum reproduces the notebook's Hartree–Fock energy, -1.116759 **Ha**, exactly ($\theta = 0$ of the UCC ansatz).

Where the exchange terms come from: the coupling H_{12}

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$H_{12} = \langle 1100|H|0011\rangle = \sum_k c_k \langle 1100|P_k|0011\rangle$. Only the four exchange strings connect the two determinants:

P_k	$P_k\,\|0011\rangle$	c_k	contribution
XXYY	$-\,\|1100\rangle$	-0.045303	$+0.045303$
XYYX	$+\,\|1100\rangle$	$+0.045303$	$+0.045303$
YXXY	$+\,\|1100\rangle$	$+0.045303$	$+0.045303$
YYXX	$-\,\|1100\rangle$	-0.045303	$+0.045303$
H_{12}			$+0.181210$

- Example: XXYY $|0011\rangle$: $X|0\rangle = |1\rangle$ twice, $Y|1\rangle = -i|0\rangle$ twice: $(-i)^2|1100\rangle = -|1100\rangle$.
- All four contributions add: $H_{12} = 4 \times 0.045303$. In second quantisation this is the exchange integral that moves an electron pair $\sigma_g^2 \to \sigma_u^2$.
- $H_{12} \neq 0$ is what makes the exact ground state differ from Hartree–Fock (correlation).

Pauli decomposition of any Hamiltonian

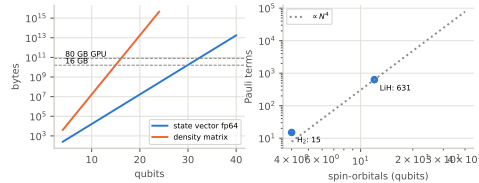
The 4^n Pauli strings form an orthogonal basis of $2^n \times 2^n$ matrices: $\mathrm{tr}(PQ) = 2^n \delta_{PQ}$. Hence

$$H = \sum_P c_P P, \qquad c_P = 2^{-n} \mathrm{tr}(PH) \in \mathbb{R}.$$

Worked example

$$\begin{pmatrix} 3 & 1-2i \\ 1+2i & -1 \end{pmatrix} = I + X + 2Y + 2Z$$

$$\mathrm{diag}(1,2,3,4) = 2.5\, II - ZI - 0.5\, IZ$$



Left: memory of the state. Right: Pauli terms grow like N^4 — H_2 15, LiH (STO-3G, 12 qubits, generated with PySCF) 631. Illustrative; LiH is not in the notebook.

Qubits, rotations and Pauli algebra

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- $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$; Born rule $P(0) = |a|^2$.
- Bloch vector $(\langle X\rangle, \langle Y\rangle, \langle Z\rangle)$.
- $R_Y(\theta) = e^{-i\theta Y/2}$: $R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$ — real amplitudes, Bloch vector $(\sin\theta, 0, \cos\theta)$.
- $e^{i\theta P} = \cos\theta\, I + i\sin\theta\, P$ for every Pauli string ($P^2 = I$).

Pauli algebra

$XY = iZ$, $YZ = iX$, $ZX = iY$; reversed order $\rightarrow$ minus sign.
Strings multiply qubit by qubit: $(X\otimes Y)(Y\otimes X) = (iZ)\otimes(-iZ) = Z\otimes Z$.
Two strings commute iff they anticommute at an *even* number of positions.

Example: $X_0X_1X_2Y_3\,|1100\rangle = i\,|0011\rangle$ (two electrons moved).

Measuring a Pauli string

Hardware measures only Z. Rotate each factor first:

factor	gates before Z-measurement
X	H
Y	$S^\dagger$, then H
Z	none
I	ignore the qubit

Each shot gives bits b; the outcome of P is $(-1)^{\sum_{k \in \mathrm{supp}\, P} b_k}$; the average estimates $\langle P \rangle$ with standard error $\leq 1/\sqrt{S}$.

Grouping

Commuting strings share an eigenbasis and can use the same shots.

- **qubit-wise commuting** (QWC): same letter or I at each position — single-qubit rotations suffice.
- H_2: all Z terms = 1 group; the 4 XY terms commute but are not QWC: 5 settings (QWC) or 2 (general).

On simulators `cudaq.observe` computes $\langle H \rangle$ exactly; with `shots_count` it samples.

Measurement grouping for H_2: a worked example

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setting	terms	basis change
none	`IIII` (constant)	—
1	10 Z-type terms	none (measure Z)
2	`XXYY`	$H, H, S^\dagger H, S^\dagger H$
3	`XYYX`	$H, S^\dagger H, S^\dagger H, H$
4	`YXXY`	$S^\dagger H, H, H, S^\dagger H$
5	`YYXX`	$S^\dagger H, S^\dagger H, H, H$

Qubit-wise commuting grouping: **5 settings**. The four exchange strings commute with each other (they differ at an even number of positions), so a joint entangling basis change measures them together: **2 settings**.

From one setting to many terms

Setting 1 returns bitstrings b; every Z-type term is evaluated from the same shots:
$\langle Z_i Z_j \rangle \approx \frac{1}{S} \sum_{\text{shots}} (-1)^{b_i \oplus b_j}$.

Larger molecules: hundreds of terms (LiH: 631), so grouping and shot allocation by $|c_k|$ matter.

Expectation values: a numerical example

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$H = 0.5\,Z_0 - 0.3\,Z_1 + 0.2\,X_0X_1$; measured $\langle Z_0\rangle = 0.6$, $\langle Z_1\rangle = -0.2$, $\langle X_0X_1\rangle = 0.5$:

$$E = 0.5(0.6) - 0.3(-0.2) + 0.2(0.5) = 0.46.$$

Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$\langle ZZ\rangle = 1$, $\langle XX\rangle = 1$, $\langle YY\rangle = -1$, $\langle X_0\rangle = \langle X_1\rangle = 0$.
For $H = -Z_0Z_1 - \frac{1}{2}(X_0 + X_1)$: $E = -1$, while $E_0 = -\sqrt{2}$.

Energy variance

$\mathrm{Var}(H) = \langle H^2\rangle - \langle H\rangle^2$ is zero exactly for eigenstates (zero-variance principle) and sets the shot noise $\sqrt{\mathrm{Var}/S}$.
$H = -Z + \frac{1}{2}X$: $H^2 = \frac{5}{4}I$; $\mathrm{Var} = 0.25$ in $|0\rangle$, 0 in the ground state.

The variational principle

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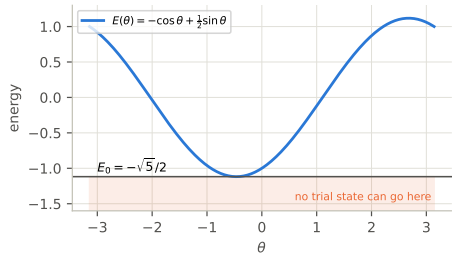
Theorem

For every normalised $|\psi\rangle$:
$E[\psi] = \langle\psi|H|\psi\rangle \geq E_0$.

Proof. Expand in the eigenbasis, $|\psi\rangle = \sum_k c_k |E_k\rangle$, $\sum |c_k|^2 = 1$:

$$\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k \geq \sum_k |c_k|^2 E_0 = E_0.$$

Equality iff $|\psi\rangle$ is a ground state. Holds for mixed states too: $\mathrm{tr}(\rho H) \geq E_0$.



One qubit, $H = -Z + \frac{1}{2}X$, trial $R_Y(\theta)|0\rangle$: every trial energy lies above E_0; the minimum touches it at $\theta^* = -\arctan\frac{1}{2}$.

Rayleigh–Ritz and what “variational” buys

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Rayleigh–Ritz

Diagonalise H inside an m-dimensional trial space: the Ritz values satisfy $\varepsilon_k \geq E_{k-1}$ (upper bounds to the lowest m exact levels). Larger spaces can only lower them.

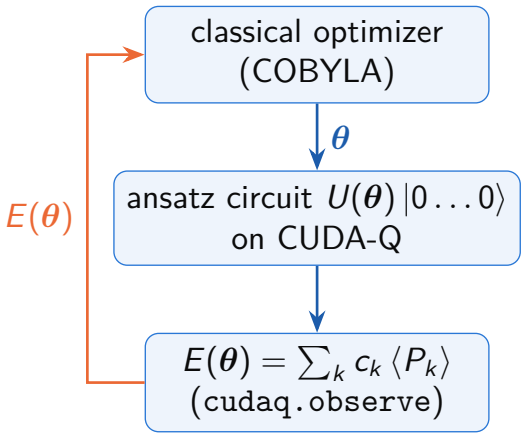
- VQE: the trial space is the set of states a circuit $U(\boldsymbol{\theta})$ can make — curved, not linear.
- An energy **below** E_0 with exact expectation values = a bug.
- With shots, single estimates can fall below by noise.

Error is linear in contamination

90 % ground state (-1.13728) + 10 % excited (-0.53821): $E = -1.0774$ Ha, error 59.9 mHa.

A good energy does not guarantee an equally good state.

The VQE loop



1. Prepare $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\ldots0\rangle$ (normalised automatically: gates are unitary).
2. Estimate $E(\boldsymbol{\theta})$ term by term.
3. The optimizer proposes new $\boldsymbol{\theta}$.
4. Repeat until converged or the budget is used.
5. Report the lowest energy seen and its angles.

Quantum part: short circuits. Classical part: optimisation. Good for noisy hardware.

Ansatz design: three families

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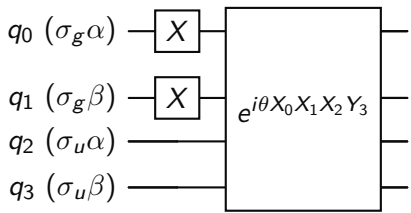
	Chemically motivated (UCC)	**Hardware-efficient (HEA)**	**Adaptive**
idea	$e^{T-T^\dagger}$ on top of Hartree–Fock	layers of native rotations + entanglers	grow the circuit operator by operator
parameters	one per excitation ($O(N^4)$ for UCCSD)	$(L+1)N$ in the notebook	few, chosen by gradients
depth	deep (Pauli gadgets, Trotter)	shallow, hardware-friendly	moderate
symmetries	conserves electron number	usually not	by construction
trainability	good start (HF), structured	barren plateaus when deep/random	good
in the notebook	`ucc_ansatz` (H_2)	`hea_ansatz` (default)	—

The ansatz decides what VQE can reach: the best VQE energy is $\min_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$ over the circuit's states, not E_0.

The Hartree–Fock reference state

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- The best single determinant: fill the lowest N_e spin-orbitals.
- Under Jordan–Wigner it is a computational basis state: $H_2 \to |1100\rangle$.
- Prepared with X gates; energy $\langle 1100|H|1100\rangle = -1.116759$ Ha (only the identity and Z terms contribute).
- It is the natural starting point ($\theta = 0$) of a chemistry VQE.



The notebook's UCC circuit: HF preparation, then one Pauli exponential (`exp_pauli`).

Unitary coupled cluster (UCC)

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$$|\psi\rangle = e^{T-T^\dagger}\,|\mathrm{HF}\rangle\,,$$
$$T = \sum_{ia} t_i^a\, a_a^\dagger a_i + \sum_{ijab} t_{ij}^{ab}\, a_a^\dagger a_b^\dagger a_j a_i$$

(i, j occupied, a, b virtual).

- $T - T^\dagger$ is anti-Hermitian $\Rightarrow$ $e^{T-T^\dagger}$ unitary (classical CC uses non-unitary e^T).
- Count (spin-orbitals): singles $n_o n_v$, doubles $\binom{n_o}{2}\binom{n_v}{2}$. LiH STO-3G: $32 + 168$; H_2: $4 + 1$.
- Under Jordan–Wigner one double excitation = 8 commuting Pauli strings, coefficients $\pm i/8$.
- Non-commuting excitations: Trotterise.

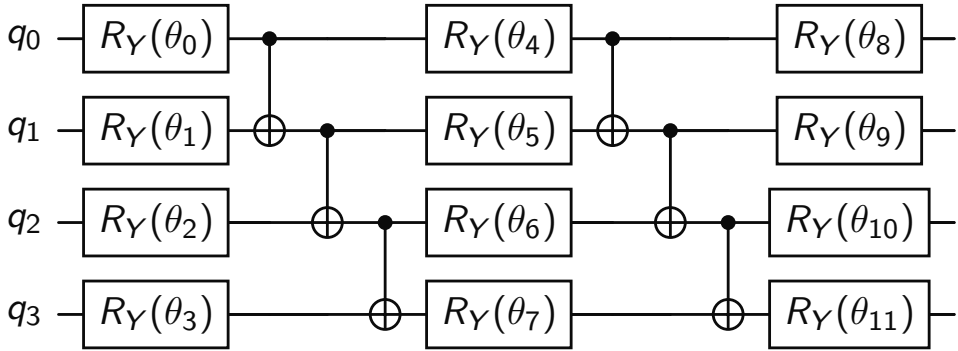
The notebook's shortcut for H_2

Keep one string: $|\psi(\theta)\rangle = e^{i\theta X_0 X_1 X_2 Y_3}\,|1100\rangle$. Since $X_0X_1X_2Y_3\,|1100\rangle = i\,|0011\rangle$,

$$|\psi(\theta)\rangle = \cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle\,.$$

Exact for H_2: the ground state lies in this plane (singles vanish by symmetry).

The hardware-efficient ansatz (default)



- L layers of [R_Y on every qubit, CNOT ladder $q \to q+1$] + final R_Y layer (drawn: $N = 4$, $L = 2$).
- Parameters $(L+1)N$, CNOTs $L(N-1)$. Default $N = 15$, $L = 2$: **45 parameters, 28 CNOTs, 73 gates** (recorded statistics, depth 20).
- Only real gates $\Rightarrow$ real amplitudes: adequate because the Ising and H_2 Hamiltonians are real symmetric.

Classical optimizers and parameter updates

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optimizer	principle (all in the notebook)
COBYLA	linear models in a trust region, derivative-free (default, `rhobeg = 0.3`)
Nelder–Mead	simplex of $n_p + 1$ points
Powell	line searches along conjugate directions
L-BFGS-B	quasi-Newton; finite-difference gradients ($n_p + 1$ runs each)

Gradients on a quantum computer

For a gate $e^{-i\theta P/2}$:
$\partial_\theta E = \frac{1}{2}[E(\theta + \frac{\pi}{2}) - E(\theta - \frac{\pi}{2})]$ — exact (**parameter shift**). HEA default: $2 \times 45 = 90$ runs per gradient.

Convergence: COBYLA stops when its trust region shrinks below `tol` or when `maxiter` is reached; the notebook keeps the best energy of the whole history.

One VQE optimisation by hand

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$H = -Z + \frac{1}{2}X$, ansatz $R_Y(\theta)\,|0\rangle$,
$E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$, start $\theta_0 = 0$.
Gradient by parameter shift, step $\theta_{k+1} = \theta_k - 0.5\,\partial_\theta E$:

k	θ_k	$E(\theta_k + \frac{\pi}{2})$, $E(\theta_k - \frac{\pi}{2})$	$\partial_\theta E$	$E(\theta_k)$
0	0	0.5, −0.5	0.500	−1.0000
1	−0.250		0.237	−1.0926
2	−0.369		0.106	−1.1130
3	−0.422		0.047	−1.1170

- Every energy is **above** $E_0 = -1.1180$ and falls monotonically towards it; $\theta_k \to \theta^* = -0.4636$.
- Each step needs 2 extra circuit runs per parameter (here: 2).
- The notebook uses COBYLA instead: no gradients, a trust region of initial size `rhobeg = 0.3`.

VQE in pseudocode

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```
input : Hamiltonian H = sum_k c_k P_k (Pauli strings), ansatz U(theta), initial angles theta0
define energy(theta):
    prepare |psi(theta)> = U(theta)|0...0> # e.g. X X on q0,q1 + exp_pauli, or RY + CNOT layers
    E = 0
    for each group of commuting P_k: # simulator: exact; hardware: basis change + shots
        E += c_k * <psi(theta)| P_k |psi(theta)>
    return E
theta* <- classical_minimise(energy, theta0) # COBYLA, MAX_EVALUATIONS = 300
report min over all evaluated energies, its angles, |E - E0| if an exact E0 is known
sample U(theta*)|0...0> to show the dominant bitstrings
```

Every call of `energy` is one full quantum simulation (or a batch of hardware shots).

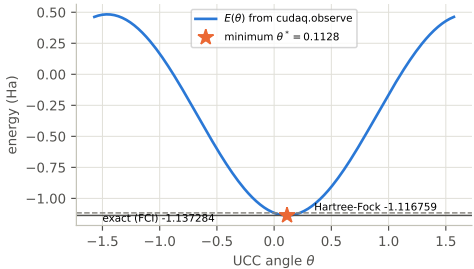
H_2: VQE as a 2×2 eigenproblem

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The UCC state lives in $\mathrm{span}\{|1100\rangle, |0011\rangle\}$.
With $H_{11} = \langle 1100|H|1100\rangle$,
$H_{22} = \langle 0011|H|0011\rangle$, $H_{12} = \langle 1100|H|0011\rangle$:

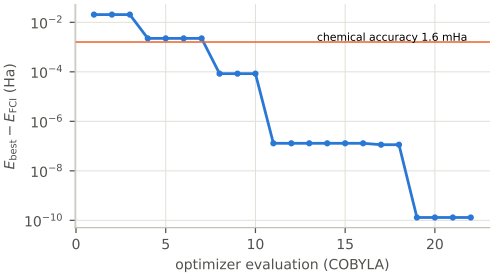
$$E(\theta) = H_{11}\cos^2\theta + H_{22}\sin^2\theta - H_{12}\sin 2\theta.$$

$H_{11} = -1.116759$, $H_{22} = 0.462618$,
$H_{12} = 0.181210$ Ha.
$E'(\theta) = 0$: $\tan 2\theta^* = \frac{2H_{12}}{H_{22}-H_{11}} = 0.2295 \Rightarrow$
$\theta^* = 0.1128$.
$E_{\min} = \frac{H_{11}+H_{22}}{2} - \sqrt{\left(\frac{H_{22}-H_{11}}{2}\right)^2 + H_{12}^2} =$
-1.137284 Ha.

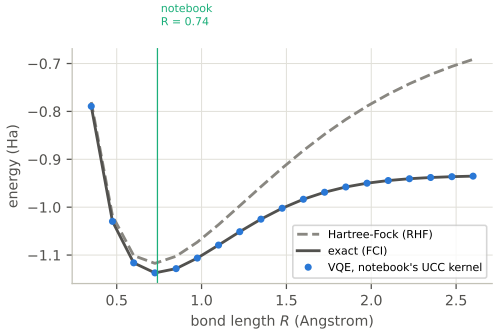


CUDA-Q: $E(\theta^*) = -1.137283834$ Ha, error $1.1e-15$ Ha vs. FCI; $P(|0011\rangle) = 0.0127$.

H_2: the VQE loop and the dissociation curve



Notebook settings (COBYLA from $\theta = 0$): chemical accuracy after 8 evaluations; best -1.1372838344 Ha after 22 evaluations (error $1.3e-10$ Ha).



Illustrative scan (PySCF + OpenFermion Hamiltonians, the notebook's UCC kernel in CUDA-Q): VQE follows FCI (max deviation $5e-15$ Ha); RHF fails at large R (244 mHa too high at 2.6 Å).

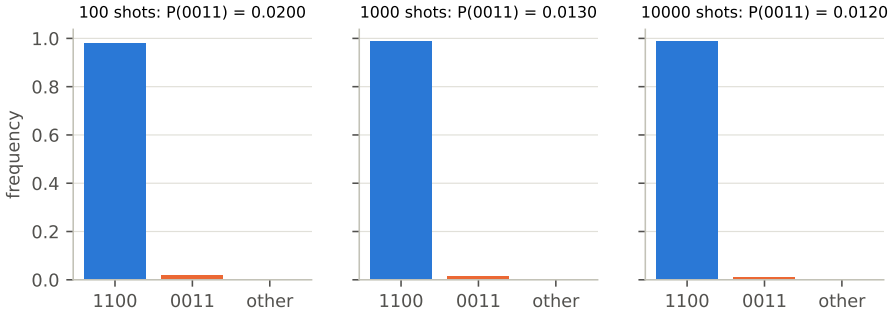
Checking the numbers with CUDA-Q: the optimal H_2 state

`cudaq.get_state(ucc_ansatz, [theta*])`, reordered to qubit 0 first, $\theta^* = 0.1128$:

basis state	amplitude	probability
$\|1100\rangle$	$+0.993647$	0.987334
$\|0011\rangle$	-0.112544	0.012666
14 others	0	0
norm		1.000000000000

- Amplitudes $\cos\theta^*$ and $-\sin\theta^*$, exactly as the formula $e^{i\theta X_0X_1X_2Y_3}\,|1100\rangle = \cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle$ predicts.
- Energy from `cudaq.observe`: -1.137283834 Ha = FCI to $1.1e-15$ Ha.
- Remember: CUDA-Q lists amplitudes with qubit 0 as the *least* significant bit; $|1100\rangle$ is index 3 there.

Measuring the H_2 VQE state: what the shots show



- Each shot returns one determinant: $|1100\rangle$ (Hartree–Fock) with probability 0.987, $|0011\rangle$ with 0.013; nothing else.
- The rare $|0011\rangle$ outcome carries the whole correlation energy (20.5 mHa). Estimating $P(0011)$ to 10 % needs $\sim 10^4$ shots — a first glimpse of VQE's measurement cost.
- The energy itself is measured from the 15 Pauli terms, not from these Z-basis counts alone.

Two-qubit Ising chain by hand

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$H = -Z_0 Z_1 - \frac{1}{2}(X_0 + X_1)$. Circuit: $R_Y(\theta_0)$, $R_Y(\theta_1)$, $\mathrm{CNOT}_{0\to 1}$.

1. $\theta_0 = \frac{\pi}{2}$, $\theta_1 = 0$: Bell state; $E = -1$.
2. Exact: H commutes with $X_0 X_1$; in the block $\{|\Phi^+\rangle, |\Psi^+\rangle\}$: $\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$, so $E_0 = -\sqrt{2}$.
3. Relative error of the Bell state: 29.3 %.
4. In general $\langle Z_0 Z_1 \rangle = \cos\theta_1$ after the CNOT (Heisenberg: CNOT maps $Z_0 Z_1 \to Z_1$).

Free-fermion reference

The notebook maps the open chain to free fermions: a $2N \times 2N$ antisymmetric matrix A, $E_0 = -\frac{1}{2}\sum |\lambda(iA)|$. $N = 2$: $-\sqrt{2}$; $N = 15$: -15.0825179883; $N = 100$: a 200×200 matrix.

The structure of `VQE.ipynb`

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1. Problem statement and mathematics
2. Imports and version check
3. **Input parameters** (the only cell to edit)
4. Hamiltonian, validation, exact reference
5. Benchmark utilities, backend selection
6. Expected output: memory, time estimate
7. Ansatz kernels, drawing, statistics, QIR export
8. **Execution**: VQE loop + final sampling, memory monitor
9. Convergence, error and bitstring plots
10. Final benchmark report (`exports/vqe/`)

Section 3 defaults

`HAMILTONIAN = "tfim"`, `N_QUBITS = 15`, `TFIM_J = 1.0`, `TFIM_H = 0.5`
`ANSATZ = "auto"` (UCC for H_2, HEA otherwise), `ANSATZ_LAYERS = 2`, `INITIAL_PARAMETER_SCALE = 0.1`
`OPTIMIZER = "COBYLA"`, `MAX_EVALUATIONS = 300`, `OBJECTIVE_SHOTS = None`
`REFERENCE_MAX_QUBITS = 18`, `SHOTS = 4000`, `SEED = 42`

Options: `"H2"` (4 qubits) or `"custom"` (list of (coefficient, Pauli word)).

Building the Hamiltonian and the exact reference (Section 4)

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```
for i in range(n_qubits - 1): # -J Z_i Z_{i+1}
    word = ["I"] * n_qubits; word[i] = word[i + 1] = "Z"
    pauli_terms.append((-float(TFIM_J), "".join(word)))
for i in range(n_qubits): # -h X_i
    word = ["I"] * n_qubits; word[i] = "X"
    pauli_terms.append((-float(TFIM_H), "".join(word)))

def build_spin_operator(terms): # sum_k c_k P_k as a CUDA-Q SpinOperator
    operator = None
    for coefficient, word in terms:
        term = coefficient * cudaq.SpinOperator.from_word(word) # character i acts on qubit i
        operator = term if operator is None else operator + term
    return operator

def tfim_exact_energy(n, J, h): # free fermions: E0 = -1/2 sum |eig(iA)|
    A = np.zeros((2 * n, 2 * n))
    ... # -h X_j and -J Z_j Z_{j+1} as Majorana couplings
    return -0.5 * float(np.sum(np.abs(np.linalg.eigvalsh(1j * A))))
```

Default: 29 terms (14 ZZ + 15 X). Reference $E_0 = -15.0825179883$; sparse diagonalisation (cross-check for $N \leq 16$) gives the same.

The ansatz kernels, line by line (Section 7a)

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```
@cudaq.kernel
def hea_ansatz(n: int, layers: int, theta: list[float]):
    qubits = cudaq.qvector(n) # |0...0>
    k = 0 # next angle
    for layer in range(layers):
        for q in range(n):
            ry(theta[k], qubits[q]); k += 1
        for q in range(n - 1):
            x.ctrl(qubits[q], qubits[q + 1])
# CNOT ladder
    for q in range(n): # final layer
        ry(theta[k], qubits[q]); k += 1

@cudaq.kernel
def ucc_ansatz(theta: list[float]):
    qubits = cudaq.qvector(4)
    x(qubits[0]); x(qubits[1]) # HF |1100>
    exp_pauli(theta[0], qubits, "XXXY")
# e^{i theta XXXY}
```

- `@cudaq.kernel`: compiled to MLIR/QIR and JIT-compiled.
- `ry(t, q)` $= e^{-itY/2}$; `x.ctrl(c, t)` = CNOT.
- Angle order: layer by layer, qubit by qubit.
- `exp_pauli(t, q, "XXXY")` $= e^{+itX_0X_1X_2Y_3}$ (sign verified in CUDA-Q 0.16): $\cos t\,|1100\rangle - \sin t\,|0011\rangle$.
- `*_sampled` variants add `mz(qubits)`.

The energy function and the optimiser (Section 8)

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```
def energy(theta):
    kernel, args = ansatz_call(kernels, theta) # HEA or UCC with its arguments
    result = cudaq.observe(kernel, hamiltonian, *args, **observe_kwargs) # exact, or shots_count=...
    value = result.expectation() # E(theta) = sum_k c_k <P_k>
    energy_history.append(value); theta_history.append(np.array(theta, dtype=float))
    return value

opt = minimize(energy, initial_theta, method="COBYLA", options={"maxiter": MAX_EVALUATIONS, "rhobeg": 0.3})
best_index = int(np.argmin(energy_history)) # best evaluated, not necessarily opt.x
best_theta, vqe_energy = theta_history[best_index], energy_history[best_index]
final_counts = cudaq.sample(kernel_s, *args_s, shots_count=SHOTS) # dominant bitstrings
```

- Printed results: optimizer message, VQE energy, reference, absolute and relative error; for H_2 "Chemical accuracy (1.6 mHa): REACHED/not reached".
- Plots: energy per evaluation and best-so-far, log error, dominant bitstrings, memory timeline.

CPU, GPU, multi-GPU and hardware

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BACKEND + SIMULATION_METHOD	CUDA-Q target	memory	executed here?
cpu + statevector	qpp-cpu	16×2^n B	**yes**
cpu + density_matrix	density-matrix-cpu	16×4^n B	no
gpu + statevector	nvidia (fp32/fp64)	8 / 16×2^n B	no (no GPU)
gpu + mps / tensornet	tensornet-mps / tensornet	entanglement-dependent	no (no GPU)
multi-gpu + statevector	nvidia, option mgpu (MPI)	split over GPUs	no
hardware	quantinuum, ionq, ...	—	no

- "auto": GPU if visible, else CPU. An explicit "gpu" without a GPU raises a RuntimeError (never touches GPU targets).
- Multi-GPU pooling needs mpiexec, one rank per GPU. Hardware returns samples only: the energy is then always estimated from shots.
- Section 6 checks memory against MEMORY_SAFETY_FRACTION and prints a rough time estimate.

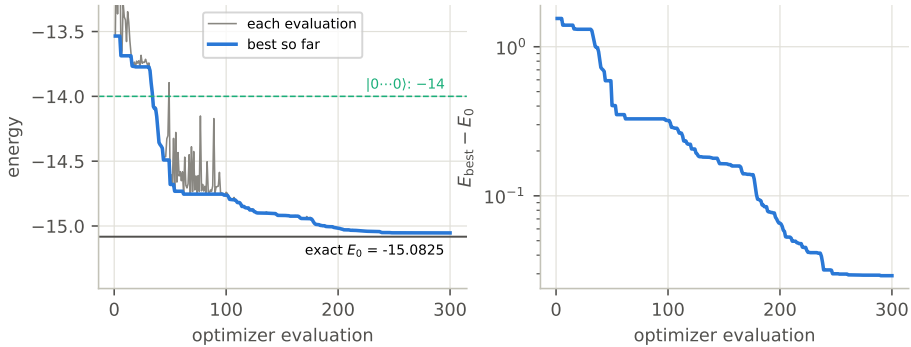
Recorded, re-executed and lecture-script results (Ising chain)

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	previously recorded	re-executed notebook	this lecture's script
machine	4 vCPU Xeon	laptop, WSL Ubuntu	laptop, WSL Ubuntu
CUDA-Q / target	0.16.0 / qpp-cpu	0.16.0 / qpp-cpu	0.16.0 / qpp-cpu
evaluations	300 (budget)	300 (budget)	300
E_{VQE}	-15.0533898284	-15.0533898284	-15.0533898284
error vs. $E_0 = -15.0825179883$	2.913×10^{-2} (0.193 %)	2.913×10^{-2} (0.193 %)	$2.913e-02$ (0.193 %)
$P(\lvert 0 \ldots 0\rangle)$ (4000 shots)	0.7005	0.7005	0.7005
execution cell / loop	70.1 s	721.6 s*	165 s

- Same seed, same initial angles, same COBYLA settings; differences in the last digits come from floating-point summation order.
- *The re-execution shared the CPU with the lecture's figure script, so its time is not a benchmark.
- Only the CPU state-vector path was executed; GPU, multi-GPU and hardware paths are implemented but **not validated**.

Convergence of the default run



- The first evaluation (-13.5348) is near the product state $|0\ldots0\rangle$ (-14); the curve approaches E_0 **from above**.
- The log-error plot is still falling at the end: COBYLA stopped because of the budget (`MAXFUN`), not because it converged.

Reading the Ising results

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Dominant bitstrings (this lecture's run)

bitstring (qubit 0 first)	probability
0000000000000000	0.7005
1000000000000000	0.0540
0000000000000001	0.0405
0000001000000000	0.0190
0001000000000000	0.0158
0100000000000000	0.0138

- **Relative error** 0.193 % looks tiny, but the product state $|0\ldots0\rangle$ already gives -14. Measured from there, VQE recovered 97.3 % of the remaining 1.0825.
- **Symmetry breaking**: the exact ground state is symmetric under flipping all spins ($P(0\ldots0) = P(1\ldots1)$); the shallow ansatz settles on one branch, at an energy cost exponentially small in N.
- Next steps: more evaluations, then more layers (the notebook suggests 4 layers / 800 evaluations at $h = J$).

Performance of the recorded run

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phase (recorded)	time
circuit construction	0.017 s
compilation (MLIR + JIT)	0.038 s
quantum execution (300 `observe`)	69.10 s
final sampling (4000 shots)	0.22 s
optimizer + Python	0.67 s
execution cell	70.09 s

- $69.10/300 = 0.23$ s per energy evaluation (73-gate circuit, 29 terms, 15 qubits).
- Pre-run estimate ~ 102.9 s: $1.5\times$ high, within the stated $\pm 3\times$.
- State memory 524 kB; peak RAM 0.92 GB — the runtime, not the state, dominates at 15 qubits.
- At 25–28 qubits each evaluation is 2^{10}–$2^{13}\times$ costlier: the notebook suggests a GPU.

Measurement shots

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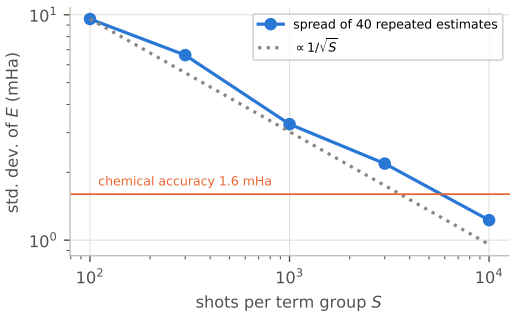
- Each estimated $\langle P_k \rangle$ has standard error $\leq 1/\sqrt{S}$; the energy combines them with weights c_k.
- Per-shot spread σ of the energy estimator $\Rightarrow S = (\sigma/\varepsilon)^2$ shots for precision ε. $\sigma = 0.1$ Ha, $\varepsilon = 1.6$ mHa: $\approx 3.9 \times 10^3$ shots *per energy evaluation*.
- Grouping commuting terms and allocating shots by $|c_k|$ reduce the cost.
- With shots, single evaluations can dip below E_0 — the notebook's text warns about this.

In the notebook

`OBJECTIVE_SHOTS = None`: exact energies (simulators). An integer: `cudaq.observe(..., shots_count=n)`. Hardware: always shots (`SHOTS`).

Try `OBJECTIVE_SHOTS = 2000`: the convergence curve becomes noisy, as on a real device.

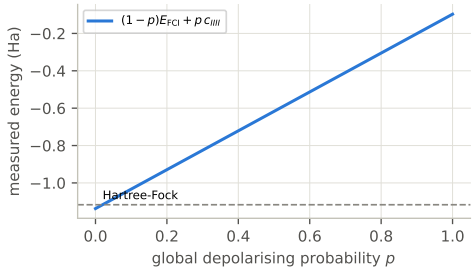
Shot noise in practice



- 40 repeated energy estimates of the optimal H_2 state with `cudaq.observe(..., shots_count=S)`.
- Spread $\approx$ 3.3 mHa at $S = 1000$ and 1.2 mHa at $S = 10\,000$: it falls like $1/\sqrt{S}$.
- Chemical accuracy needs the spread well below 1.6 mHa — for every energy evaluation of the optimisation loop.
- Single estimates may fall *below* the FCI energy; only the average obeys the variational bound.

Noise, error mitigation and hardware

- Noise turns pure states into mixtures; the variational bound still holds ($\mathrm{tr}\,\rho H \geq E_0$), but energies rise.
- Complete depolarisation: $E = \mathrm{tr}\,H/2^n = c_{I\ldots I} = -0.0971$ Ha for H_2.
- Illustrative global-depolarising model: $E(p) = (1-p)E_{\mathrm{FCI}} + p\,c_{IIII}$; already $p = 0.020$ loses the whole correlation energy.
- **Error mitigation**: zero-noise extrapolation ($E(\lambda{=}1) = -1.10$, $E(3) = -1.00 \Rightarrow$ $E(0) \approx -1.15$; no variational guarantee), readout correction, symmetry verification (electron number).
- **Hardware**: limited connectivity (SWAPs), native gates, CNOT errors 10^{-3}–10^{-2}: shallow ansätze preferred.



Model for illustration, not a hardware measurement; `Noise_Simulation.ipynb` shows CUDA-Q noise channels.

Limitations

- **Measurement cost**: $O(N^4)$ terms, $(\sigma/\varepsilon)^2$ shots each evaluation, hundreds of evaluations.
- **Barren plateaus**: for deep random (hardware-efficient) circuits the gradient variance decays exponentially with n.
- **Local minima** and optimizer budgets (the default run stopped at its budget).
- **Expressiveness**: the result is only as good as the ansatz; shallow HEA misses correlations, deep UCC is long.
- **Noise** raises energies; mitigation costs more shots.

Honest status

No quantum advantage for chemistry has been demonstrated with VQE. Classical methods (CCSD(T), DMRG) handle molecules far beyond today's devices. VQE's value today: a well-understood hybrid algorithm and a benchmark; the notebook reproduces small problems exactly on a simulator.

VQE compared with classical methods and with phase estimation

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method	cost	quality
Hartree–Fock	N^4	no correlation (H_2: -20.5 mHa missed)
CCSD(T)	N^7	"gold standard" near equilibrium; fails for bond breaking
DMRG	poly. for 1D	excellent for chains (like the Ising model)
FCI	$\binom{N}{N_e}$	exact in the basis; small molecules only
VQE	shots $\times$ evaluations	variational upper bound; limited by ansatz, shots, noise

VQE versus quantum phase estimation (QPE)

- QPE (see the QFT lecture) reads E from the phase of e^{-iHt}: precision ε with $O(1/\varepsilon)$ evolution time, deep circuits, fault tolerance needed.
- VQE: shallow circuits, but $O(1/\varepsilon^2)$ shots and a classical optimiser.
- VQE states can serve as good initial states for QPE.

Excited states with VQE

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Variational quantum deflation (VQD)

After finding $|\psi_0\rangle$, minimise

$$F(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|H|\psi(\boldsymbol{\theta})\rangle + \beta\,|\,\langle\psi_0|\psi(\boldsymbol{\theta})\rangle\,|^2$$

with β larger than the gap $E_1 - E_0$: the penalty pushes the new state away from the ground state, so the minimum is E_1.

The overlap $|\,\langle\psi_0|\psi\rangle\,|^2$ is measured with a swap test or by running $U(\boldsymbol{\theta}_0)^\dagger U(\boldsymbol{\theta})$ and counting $|0\ldots0\rangle$.

- **Subspace-search VQE**: optimise several orthogonal input states together with one circuit.
- **Quantum subspace expansion**: measure H in a small subspace around the VQE state and diagonalise classically (Rayleigh–Ritz).
- Excitation energies give spectra (colours, absorption) — not covered by the notebook.

Advanced VQE and applications

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Better ansätze and fewer qubits

- **ADAPT-VQE**: add operators one by one from a pool, chosen by energy gradient.
- **k-UpCCGSD**, qubit-coupled cluster: cheaper chemistry-inspired circuits.
- **Active spaces**, frozen core (LiH: 12 $\rightarrow$ 10 qubits), tapering.

Excited states

Variational quantum deflation (penalise overlap with found states), subspace-search VQE, quantum subspace expansion.

Applications

- reaction energies and barriers; catalysts
- potential energy surfaces (bond breaking)
- battery and photovoltaic materials (lattice models, Hubbard)
- magnetism: spin chains such as the notebook's Ising model

None of the advanced variants is implemented in `VQE.ipynb`.

Try it yourself: `VQE.ipynb`

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1. `pip install -r requirements.txt`, open `VQE.ipynb`, run all cells.
2. Edit only Section 3:
   - `HAMILTONIAN = "H2"`: UCC, chemical accuracy in a few evaluations
   - `TFIM_H = 1.0`: the critical point, much harder
   - `ANSATZ_LAYERS = 4`, `MAX_EVALUATIONS = 800`

Experiments

- `OBJECTIVE_SHOTS = 2000`: optimise with shot noise
- `OPTIMIZER = "L-BFGS-B"`: gradient-based
- `HAMILTONIAN = "custom"` with your own Pauli words
- `N_QUBITS = 24`, `BACKEND = "gpu"`: timing on a GPU

Common mistakes and how to spot them

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- **Energy below E_0** with exact expectation values: a bug (sign, missing term, reference), never a success.
- **Amplitude vs. probability**: $\cos\frac{\theta}{2}$ is an amplitude; $P = \cos^2\frac{\theta}{2}$.
- **Bit order**: notebook strings are qubit 0 first; CUDA-Q state vectors are qubit 0 least significant.
- **Half angles**: $R_Y(\theta) = e^{-i\theta Y/2}$, but `exp_pauli(theta, ...)` $= e^{+i\theta P}$ (no $\frac{1}{2}$, plus sign).
- **Forgetting the constant $c_{I\ldots I}$** (nuclear repulsion) when comparing with total energies.
- **Measuring X or Y without a basis change**.
- **Relative error looks small**: compare with a trivial baseline (Ising: $|0\ldots0\rangle$ already gives -14).
- **Budget mistaken for convergence**: read the optimizer's stopping message (`MAXFUN`).

Exercises

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1. Eigenvalues and ground-state vector of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; Rayleigh quotient of $(1, 0)$.
2. Decompose $\mathrm{diag}(2, 0, 0, -2)$ into Pauli strings.
3. Write $a_1^\dagger$ on 3 qubits under Jordan–Wigner.
4. Compute $X_0 Y_1 |00\rangle$ and $\langle 00| X_0 X_1 |00\rangle$.
5. For $H = Z + X$ and trial $R_Y(\theta)|0\rangle$: $E(\theta)$, θ^* and E_0.
6. Parameters, CNOTs and gates of the notebook's HEA for $N = 6$, $L = 3$.
7. An energy estimator has $\sigma = 0.05$ Ha. Shots for a standard error of 1 mHa?
8. Why can a VQE energy computed with exact expectation values never be below E_0?

Answers

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1. ± 1; ground state $\frac{1}{\sqrt{2}}(1,-1)$; $R((1,0)) = 0 \geq -1$.
2. $\mathrm{diag}(2,0,0,-2) = Z_0 + Z_1$ (check $|11\rangle$: $-1-1=-2$).
3. $a_1^\dagger = Z_0 \, \frac{1}{2}(X_1 - iY_1) \otimes I_2$.
4. $X_0 Y_1 \, |00\rangle = |1\rangle \otimes i\,|1\rangle = i\,|11\rangle$; $\langle 00|X_0 X_1|00\rangle = \langle 00|11\rangle = 0$.
5. $E = \cos\theta + \sin\theta = \sqrt{2}\sin(\theta + \frac{\pi}{4})$; $\theta^* = -\frac{3\pi}{4}$; $E_0 = -\sqrt{2}$.
6. $(3+1)\cdot 6 = 24$ parameters, $3 \cdot 5 = 15$ CNOTs, 39 gates.
7. $S = (0.05/0.001)^2 = 2500$.
8. The circuit state is normalised and $\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k \geq E_0$ (variational principle).

Summary

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- Ground-state energies are eigenvalue problems of exponentially large Hermitian matrices.
- Molecules: Born–Oppenheimer $\rightarrow$ second quantisation $\rightarrow$ Jordan–Wigner/Bravyi–Kitaev $\rightarrow$ $H = \sum_k c_k P_k$ with $O(N^4)$ terms.
- **Variational principle**: $\langle\psi|H|\psi\rangle \geq E_0$ — VQE minimises the energy of a circuit state.
- Ansätze: Hartree–Fock + UCC (exact for H_2 with one parameter), or hardware-efficient R_Y + CNOT layers.
- Energies are measured term by term; shots and noise dominate the cost on hardware.
- `VQE.ipynb`: Ising chain within 0.193 % of the exact E_0 (budget-limited); H_2 reaches FCI — reproduced here with CUDA-Q.

Further reading

- A. Peruzzo et al., *A variational eigenvalue solver on a photonic quantum processor*, Nat. Commun. 5 (2014)
- J. McClean et al., *The theory of variational hybrid quantum-classical algorithms*, NJP 18 (2016)
- A. Kandala et al., *Hardware-efficient VQE for small molecules*, Nature 549 (2017)
- J. Tilly et al., *The Variational Quantum Eigensolver: a review*, Phys. Rep. 986 (2022)
- NVIDIA CUDA-Q documentation
- This repository: `VQE.ipynb`, `assessments/VQE_*.pdf`